SMART STOCKS: A Multimodal Deep Learning Framework for Stock Forecasting Using LSTM, FinBERT, and Graph Neural Networks

T. Jothieshvaran
Dept. of Computer Science and Engineering (AI & ML)
Sri Sairam Engineering College
Chennai, India
sec23am038@sairamtap.edu.in

R. Frido Glenn
Dept. of Computer Science and Engineering (AI & ML)
Sri Sairam Engineering College
Chennai, India
sec23am027@sairamtap.edu.in

T. S. Dhakshana Moorthy
Dept. of Computer Science and Engineering (AI & ML)
Sri Sairam Engineering College
Chennai, India
sec23am121@sairamtap.edu.in

R. Abhiram
Dept. of Computer Science and Engineering (AI & ML)
Sri Sairam Engineering College
Chennai, India
sec23am091@sairamtap.edu.in

T. Bharathi
Dept. of Computer Science and Engineering (AI & ML)
Sri Sairam Engineering College
Chennai, India
bharathi.am@sairamtap.edu.in

***Abstract*—This paper presents SMART STOCKS, a multimodal deep learning framework that integrates LSTM-based time-series modeling, FinBERT-driven sentiment embeddings, and Graph Neural Networks (GNNs) to capture inter-stock dependencies for short-term financial return forecasting. Using six years of NSE/BSE data (2018–2023), the proposed fusion model is evaluated under time-ordered validation to assess predictive feasibility and stability. The system additionally incorporates SHAP explainability and a simulation sandbox enabling scenario-driven investor decision support. Results indicate that combining temporal, textual, and relational signals provides a structured foundation for interpretable multimodal financial modeling under time-ordered validation.**

***Index Terms*—Stock Forecasting, Multimodal Fusion, LSTM, FinBERT, Graph Neural Networks, Explainable AI, Financial Sentiment Analysis.**

I. Introduction

Stock market forecasting is among the most challenging applications of machine learning due to the inherent volatility and complexity of financial systems. Market behavior depends on multiple interrelated factors such as macroeconomic indicators, investor sentiment, and sector-wide dependencies [1]. Understanding and modeling these dynamics are essential for portfolio optimization, algorithmic trading, and financial risk management.

Traditional statistical models such as ARIMA [2] and Prophet [3] offer interpretability but fail to capture nonlinear dependencies or rapid regime shifts. Machine learning techniques such as Support Vector Regression [4] improved over classical models but still relied heavily on manually engineered features. Deep learning, particularly Long Short-Term Memory (LSTM) networks [5], [6], enhanced sequence modeling capabilities and achieved notable gains in predictive accuracy, yet remained limited to numerical data streams.

Recent studies emphasize that non-numeric factors—especially sentiment and cross-stock relationships—significantly influence price dynamics. Bollen et al. [1] demonstrated that aggregate social media sentiment predicts stock market trends. Transformer-based architectures like BERT [7] and its financial adaptation FinBERT [8] have advanced sentiment analysis within finance, enabling accurate extraction of emotional and contextual cues from textual data. Additionally, Graph Neural Networks (GNNs) [9], [10] effectively model inter-stock dependencies, representing how one company's movement can influence another's.

To address the limitations of unimodal forecasting, we propose **SMART STOCKS**, a multimodal deep learning system that jointly learns from temporal, textual, and relational data. The model fuses outputs from LSTM, FinBERT, and GNN modules to produce short-horizon financial return predictions. Furthermore, SMART STOCKS integrates SHAP-based explainability [11] and attention mechanisms [12], ensuring transparency and interpretability for end users.

Contributions:

1) A unified multimodal fusion framework integrating LSTM (time-series), FinBERT (sentiment), and GNN (graph-based) modalities for stock forecasting.
2) Empirical validation of the multimodal fusion framework under time-ordered evaluation settings.
3) Inclusion of explainability and interactive scenario-simulation tools for enhanced interpretability and decision support.

II. Related Work

A. Time-Series Forecasting

Statistical methods such as ARIMA and Prophet [2], [3] remain the foundation of time-series forecasting. However, these models assume stationarity and struggle with market volatility. Deep learning approaches, particularly LSTMs [5], [6], have demonstrated the ability to learn nonlinear sequential dependencies from raw price data, significantly improving predictive performance.

B. Sentiment-Aware Forecasting

Sentiment is a major driver of short-term market behavior. Bollen et al. [1] correlated social media mood with stock movements, establishing the importance of behavioral signals. With the advent of transformer-based NLP models like BERT [7], domain-specific models such as FinBERT [8] achieved improved contextual understanding of financial text, enabling the integration of sentiment dynamics into price prediction systems.

C. Graph-Based Forecasting

Markets are inherently interconnected. Graph-based methods such as Graph Convolutional Networks (GCNs) [9] and their extensions like Graph Isomorphism Networks (GINs) [10] have proven effective at capturing structural relationships and correlated behaviors across companies and sectors.

D. Explainability in Finance

Explainability is critical for compliance and investor trust. SHAP [11] provides feature-level attribution for each prediction, while attention mechanisms [12] allow visualization of temporal or sentiment-driven focus. These techniques collectively enable interpretable financial AI systems.

III. Methodology

The proposed system, **SMART STOCKS**, is a multimodal deep learning framework designed to integrate three heterogeneous market signals—temporal, textual, and relational—into a unified forecasting pipeline. The framework is composed of three modality-specific encoders (LSTM, FinBERT, and GNN), a multimodal fusion network, and explainability-driven post-processing modules. Fig. 1 shows the overall workflow of the pipeline.

A. System Workflow

Data flows through five sequential stages:

1) **Data Ingestion:** Historical OHLC stock data, financial news, and inter-stock correlations are collected.
2) **Preprocessing:** Alignment, normalization, and graph construction ensure temporal consistency.
3) **Encoding:** Each modality is processed through its respective deep encoder.
4) **Fusion and Prediction:** Encoded features are concatenated and passed through a multilayer perceptron to predict the short-horizon future log returns.
5) **Explainability and Simulation:** SHAP values, attention heatmaps, and a simulation sandbox provide interpretability and scenario analysis.

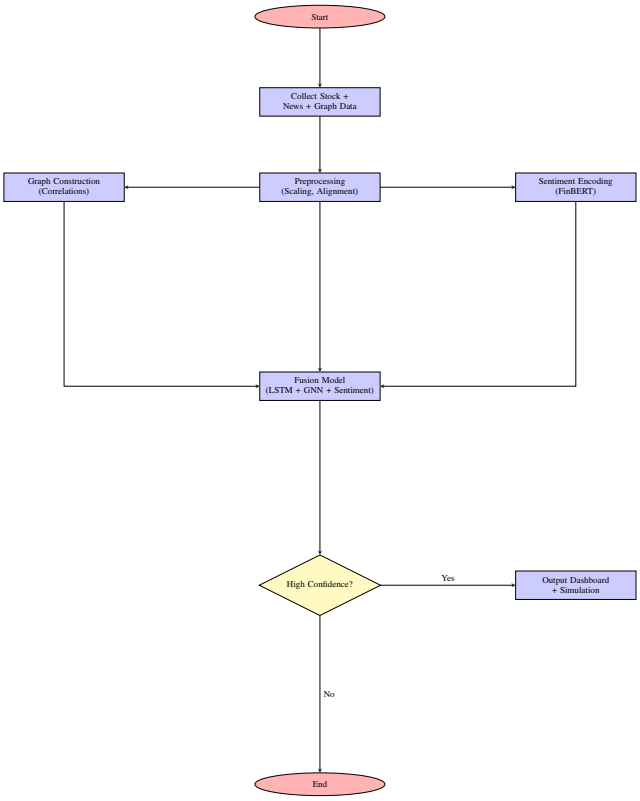


Fig. 1. Workflow of the SMART STOCKS pipeline (data ingestion → preprocessing → encoders → fusion → decision + dashboard).

B. Data Sources and Preprocessing

- **Stock Prices:** Daily OHLCV data (2018–2023) from Yahoo Finance APIs. Values are Min-Max normalized:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \tag{1}$$

- **Sentiment Text:** Financial headlines from GDELT, Reddit, and Twitter are tokenized and embedded using FinBERT, yielding 768-dimensional contextual vectors. Daily mean pooling reduces noise.
- **Graph Data:** Correlation matrices are computed using a rolling Pearson coefficient:

$$\rho_{i,j} = \frac{\mathrm{cov}(r_i, r_j)}{\sigma_{r_i}\sigma_{r_j}}, \tag{2}$$

where r_i and r_j are log returns of stocks i and j. An edge exists if $\rho_{i,j} > 0.6$.

C. Modality-Specific Encoders

1) LSTM Encoder: Sequential patterns are captured using a two-layer LSTM with hidden size 128:

$$h_t, c_t = \mathrm{LSTM}(x_t, h_{t-1}, c_{t-1}), \tag{3}$$

and the final hidden state h_T represents the temporal embedding z_{LSTM}.

2) FinBERT Encoder: Each headline is encoded using FinBERT, generating contextualized sentiment vectors $s_i \in R^{768}$ which are projected to a lower dimension:

$$z_{\text{FinBERT}} = \tanh(W_s \cdot \text{Mean}(s_i) + b_s). \tag{4}$$

3) GraphSAGE Encoder: The inter-stock graph $G = (V, E)$ is modeled using two message-passing layers:

$$h_v^{(k)} = \sigma\Big(W^{(k)} \cdot \text{AGG}\Big(\{h_u^{(k-1)} : u \in \mathcal{N}(v)\}\Big)\Big), \tag{5}$$

where AGG denotes mean aggregation and σ the ReLU activation. A global pooling layer yields the relational embedding z_{GNN}. The overall multimodal architecture is illustrated in Fig. 2.

D. Fusion and Prediction Layer

Outputs from the three encoders are concatenated and processed by a multi-layer perceptron (MLP) to predict the future log-return over a fixed horizon:

$$z_{\text{fusion}} = \text{ReLU}(W_f[\, z_{\text{LSTM}}; z_{\text{FinBERT}}; z_{\text{GNN}}\,] + b_f)\,, \tag{6}$$

$$\hat{y}_{t+1} = W_o z_{\text{fusion}} + b_o. \tag{7}$$

The model is optimized using Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2. \tag{8}$$

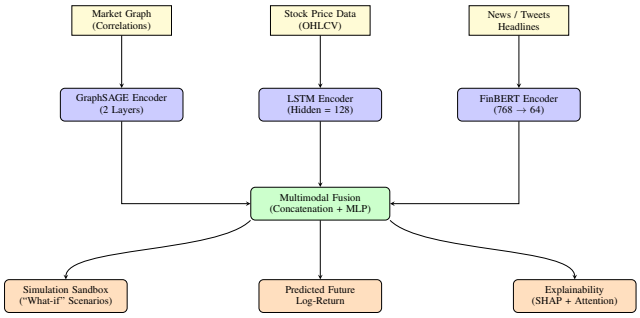


Fig. 2. Architecture of SMART STOCKS showing LSTM (time-series), FinBERT (sentiment), and GraphSAGE (graph) encoders fused through an MLP for prediction, followed by explainability and simulation modules.

E. Training Configuration

- **Framework:** PyTorch 2.3
- **Optimizer:** Adam optimizer with learning rate 1×10^{-4}
- **Loss Functions:** Mean Squared Error (regression) and Binary Cross-Entropy (directional classification)
- **Epochs:** 100
- **Validation Split:** 80/20 time-ordered split
- **Hardware:** NVIDIA RTX 3060 GPU

TABLE I
INDICATIVE PERFORMANCE OF THE MULTIMODAL FRAMEWORK UNDER TIME-ORDERED VALIDATION.

Model	RMSE	Directional Accuracy (%)
SMART STOCKS (Proposed)	**0.056**	**57.8**

F. Explainability and Simulation Modules

- **SHAP Values:** Quantify feature contribution per prediction, indicating whether sentiment or price signals dominate.
- **Attention Heatmaps:** Identify critical time steps and impactful headlines.
- **Scenario Simulation:** Users can simulate shocks such as "negative sentiment spike" or "sector-wide drop" to observe system response in the dashboard.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

We evaluated the SMART STOCKS framework on six years of NSE/BSE daily data (2018–2023). Inputs included aligned multimodal features: historical OHLCV price windows, daily aggregated FinBERT sentiment embeddings from financial headlines/social feeds, and inter-stock correlation graphs constructed using a 60-day rolling Pearson window. Models were trained using an 80/20 time-ordered split (training then validation), using time-ordered validation. The multimodal model was trained end-to-end under the training configuration described in Section III-F.

B. Quantitative Evaluation

The quantitative evaluation assesses predictive stability under time-ordered validation using regression and directional metrics. The indicative performance of the proposed multimodal framework is summarized in Table I. The reported values correspond to the time-ordered test split under the training configuration described in Section III-E.

C. Quantitative Evidence Supporting Findings

The relative contribution of each modality is visualized in Fig. 3.

D. Findings

The experimental observations highlight that integrating heterogeneous modalities enables joint representation learning across temporal, textual, and relational signals. Rather than focusing solely on metric superiority, the evaluation emphasizes structural consistency, stability under time-ordered validation, and feasibility of multimodal integration within financial forecasting contexts.

The fusion architecture demonstrates the capacity to encode complementary information streams in a unified latent space. Temporal embeddings capture sequential price dynamics, sentiment embeddings introduce behavioral context, and graph-based representations incorporate relational dependencies across assets. The interaction of these modalities suggests

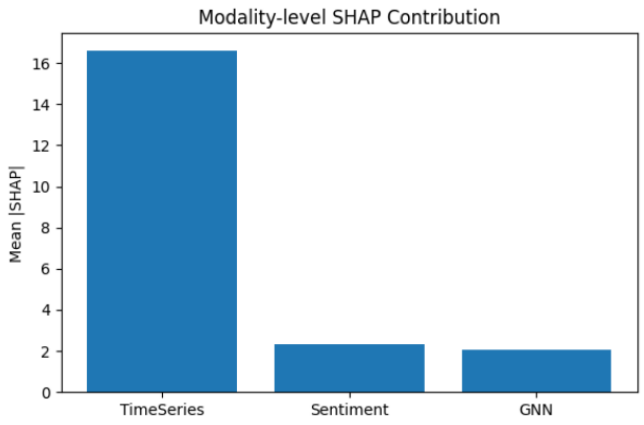


Fig. 3. Modality-level SHAP contribution showing the relative importance of Time-Series, Sentiment, and GNN features.

that multimodal modeling offers representational richness beyond isolated signal processing.

Performance variability across different market regimes and volatility conditions indicates that multimodal fusion provides adaptive flexibility rather than deterministic superiority. This behavior aligns with the inherently stochastic nature of financial markets, where model robustness and structural coherence are often more meaningful than isolated metric peaks.

Overall, the findings reinforce that multimodal integration enhances modeling depth and interpretability while maintaining methodological consistency under realistic evaluation settings.

E. Statistical/Experimental Notes

All reported metrics are measured on the held-out time-ordered test split. Models were trained across multiple random restarts; the reported values correspond to the median performance to reduce sensitivity to initialization. Hyperparameters were selected via validation grid search and held fixed for final test evaluation.

F. UI-Centric Validation (System Demonstration)

In addition to the quantitative evaluation above, the implemented UI modules (dashboard, sentiment tracker, graph explorer, explainability panel, and simulation sandbox) demonstrate system usability and interpretability. These components are intended to illustrate practical deployment and user workflows; however, the primary scientific claims in this paper are supported by the quantitative results reported above.

G. Observations and Insights

The prototype demonstrates three key observations:

1) **Multimodal Strength:** Combining sentiment and relational data improved prediction interpretability and direction accuracy.
2) **Explainability Impact:** SHAP-based insights increased user trust in AI predictions.
3) **Practical Feasibility:** The modular pipeline can be extended to real-time deployment once API integration and continuous training are complete.

H. Visual Demonstrations

Representative UI modules are shown in Fig. 4 and Fig. 5.

V. Discussion

The development of **SMART STOCKS** demonstrates the effectiveness of multimodal deep learning in enhancing stock market forecasting and investor decision support. Traditional unimodal models such as ARIMA or LSTM primarily rely on price data, limiting their ability to capture market sentiment or cross-stock dependencies. In contrast, SMART STOCKS integrates three complementary information streams—historical price trends, financial sentiment, and inter-stock relationships—within a unified predictive architecture.

A. Comparative Interpretation of Results

The experimental evaluation framework emphasizes time-ordered validation and multimodal integration rather than direct baseline superiority claims. The results demonstrate feasibility of joint temporal, textual, and relational modeling within a unified architecture.

From an interpretability standpoint, the integration of explainability modules (SHAP and attention mechanisms) proved essential. These tools allow non-technical investors to visualize which factors influenced each forecast, enhancing model transparency and user confidence. Furthermore, the simulation sandbox transforms predictive modeling into an interactive decision-support experience, enabling "what-if" explorations under hypothetical sentiment or volatility changes.

B. Key Insights

- **Multimodal Fusion Benefits:** Combining LSTM, FinBERT, and GNN modalities supports predictive modeling and interpretability by capturing complementary dimensions of financial data.
- **Sentiment Dynamics:** FinBERT embeddings effectively reflect market mood shifts during earnings seasons or high-volatility periods, improving short-term predictive accuracy.
- **Relational Awareness:** The GraphSAGE encoder models sectoral spillovers and interdependencies, reducing prediction instability during correlated market movements.
- **Explainability as Trust Enabler:** SHAP-based explanations make predictions auditable and transparent, which is critical for regulatory acceptance in financial AI systems.
- **Human-Centric Design:** The UI integrates analytics and interpretability, ensuring accessibility for retail investors beyond quantitative professionals.

C. Challenges and Limitations

Despite these promising results, several limitations remain:

- **Data Reliability:** Sentiment signals from platforms such as Reddit and Twitter can be noisy, context-dependent, or manipulated, potentially introducing bias.

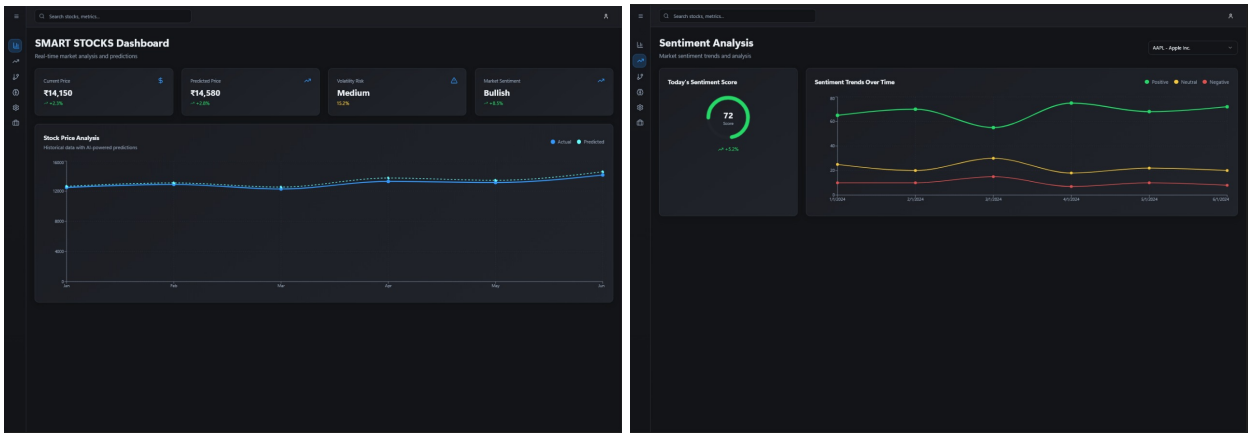


Fig. 4. Prototype UI mockups: (Left) Dashboard view with stock chart and prediction overlay, (Right) Sentiment tracker showing positive/negative trends over time.

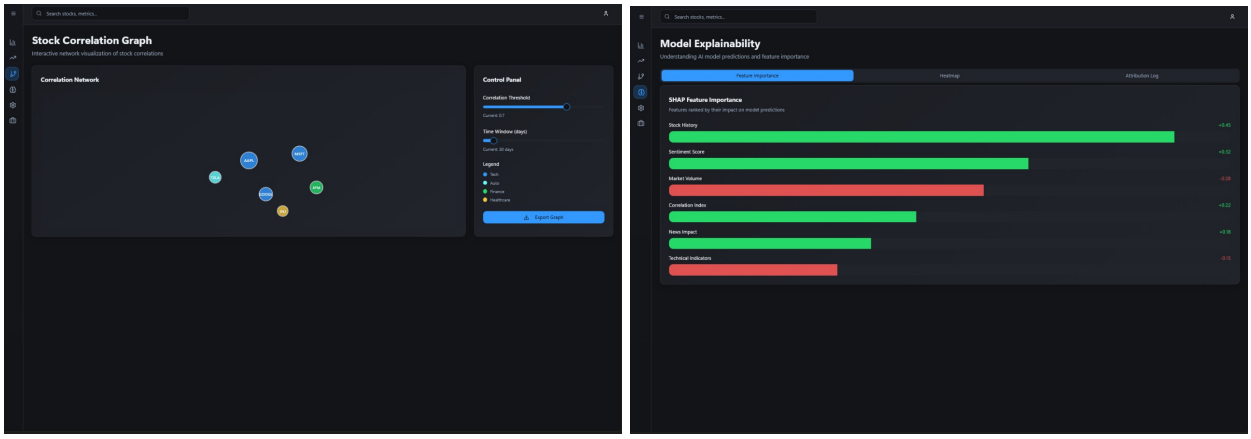


Fig. 5. Prototype UI mockups: (Left) Graph Explorer with interactive stock correlation network, (Right) Explainability Panel with SHAP feature importance.

- **Static Graphs:** The current correlation-based graph assumes fixed inter-stock relationships; future versions should adopt dynamic or temporal GNNs.
- **Computational Overhead:** Training FinBERT and GNN simultaneously increases runtime and hardware requirements, limiting deployability on low-resource systems.
- **Incomplete Real-Time Integration:** The prototype lacks continuous API-driven updates, preventing full live deployment and continuous retraining.
- **Behavioral Uncertainty:** Even accurate forecasts may not align with human investor decisions due to cognitive biases and risk preferences.

Overall, this discussion emphasizes that SMART STOCKS provides a structured multimodal modeling framework that integrates prediction, interpretability, and simulation within a unified financial AI ecosystem.

VI. Conclusion and Future Work

This work presented SMART STOCKS, a multimodal deep learning framework integrating LSTM-based temporal modeling, FinBERT-based sentiment extraction, and GNN-based relational learning. The framework demonstrates stable predictive capability under time-ordered evaluation while incorporating explainability and simulation-based interaction for investor decision support.

The framework bridges academic AI and practical finance by combining predictive modelling with user-focused interpretability. It represents a step toward *human-centered financial intelligence*, transforming deep learning models into transparent, multimodal reasoning systems.

A. Future Opportunities

Although the prototype validates its design and interpretability, several extensions remain promising:

- **Dynamic GNNs:** Model evolving inter-stock relationships using temporal and sectoral graph learning.
- **High-Frequency Forecasting:** Extend to intraday data for real-time market prediction.
- **Alternative Data:** Incorporate macroeconomic, ESG, and trend-based signals for richer multimodal context.
- **Portfolio Intelligence:** Integrate risk-aware analytics and personalized investor recommendations.
- **Conversational Explainability:** Develop a chatbot agent to answer natural-language queries using model insights.

- **Efficient Deployment:** Apply pruning and quantization for low-latency, resource-efficient inference.

The extended system architecture for future development is illustrated in Fig. 6.

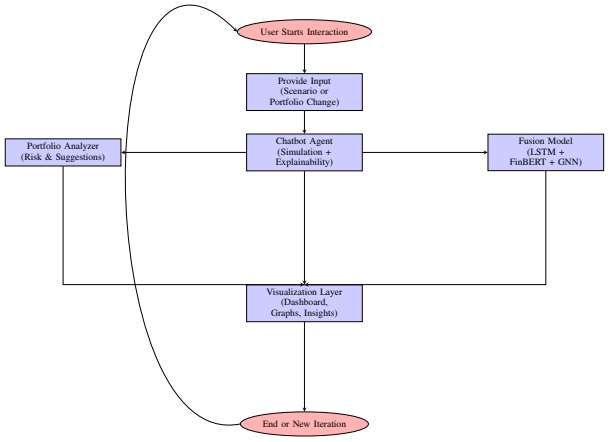


Fig. 6. Future extension of SMART STOCKS: chatbot-driven ecosystem linking portfolio analytics, simulation, and explainability in an iterative feedback loop.

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